ScreenShot Management (SSM)

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DEFINITION

This format allows an emulator to generate screenshots on demand of the executable code.

The principle consists in using a "*neutral*" opcode not used by the Z80A but which will be identified by an emulator. This management can be conditional (or compiled on purpose during tests).

There are 79 Z80A instructions prefixed by #ED.
177 (e.g. 256-70) "neutral" non-active codes can therefore be used after an "#ED" prefix code.

The principle of SSM is as follows: In memory, two #ED codes suffixed by a value define a code. With 177 possibilities per code, this represents 31,329 different combinations.

The following sequence of bytes represents an SSM request for the emulator: **#ED <#LL> #ED <#HH>** (# indicates a hexadecimal value).

<#LL> and <#HH> can each be :
- between #00 and #3F (included)
- between #7F and #9F (included)
- between #A4 and #A7 (included)
- between #AC and #AF (included)
- between #B4 and #B7 (included)- between #BC and #BF (included)
- between #C0 and #FD (included)

When the Z80A emulator processes an SSM request:

- It collects the 2 values <#LL> and <#HH>. SSM Code value = #HH x 256 + #LL
- An SSM code is recognized when 2 consecutive series of #ED #xx are encountered. For example, with the byte sequence ED 3F **00** ED **3E** ED **3D** then LL code is **3E** and the HH code is **3D**. The 00 (or any other code <> ED) resets the expectation of the 16-bit SSM code.

SSM CODES

SSM codes can have several functions within an emulator.

Screenshot Generation :

When SSM handling is enabled in an emulator, it can use the read SSM code to generate a screenshot immediately after reading the **#HH** byte.
Some SSM codes are reserved for other (or later) uses: Code #0000 and all #FFxx codes are reserved (this represents 178 values).

Image name format suggested (especially if the screenshots are intended for the SHAKERLAND portal):
<Emulator name>_<CRTC number>_<HHLL code>.<image extension>

The format of the image (and therefore its extension) depends solely on the emulator.
Examples:
CRTC 2, ED E3 ED 02 with AMSPIRIT: **AMSPIRIT_2_02E3.bmp**
CRTC 0, ED DA ED 01 with SUGARBOX: **SUGARBOX_0_01DA.jpg**

SSM Code	Definition
#0000	**#ED #00 #ED #00** is a specific SSM code, used to break a **wait_ssm0000** instruction in a CSL file. See definition of a CSL file.
#FFFF	If the emulator interprets the SSM **#FFFF** code **(#ED #FF #ED #FF**), then a snapshot is saved with the name defined by the "**snapshot_name**" instruction of a CSL file. See definition of a CSL file. If the name has not been defined by CSL, then the naming follows the rule suggested in the previous chapter and the code is then FFFF and the extension is ".sna"
#FFFE	If the emulator interprets the SSM **#FFFE** code **(#ED #FE #ED #FF**), then a screenshot is saved with the name defined by the "**screenshot_name**" instruction of a CSL file. See definition of a CSL file. If the name has not been defined by CSL, then the naming follows the rule suggested in the previous chapter and the code is then FFFE.
#FFFD	If the emulator interprets the SSM code **#FFFD (#ED, #FD, #ED, #FF**), then this allows the **Sikoview** code inspector to determine the starting point for event logging.
#FFFC	If the emulator interprets the SSM code **#FFFC (#ED, #FC, #ED, #FF**), then this allows the **Sikoview** code inspector to determine the event logging breakpoint.
	The other 173 available codes are reserved for future use.

DOCUMENTATION

SSM & CSL documents & files are available at: https://shaker.logonsystem.eu

From version 2.5, SHAKER integrates SSM codes.